

Grade 10 ICT – Information and Communication Technology (ICT) Notes

1. Data and Information

Data

Data are raw facts and figures that have not been processed.

Examples:

- Student names
- Subject marks
- Numbers
- Images
- Symbols

Information

Information is meaningful data obtained after processing.

Examples:

- Total marks
- Average marks
- Student rank
- Reports

Example

Name	Maths	Science	Total	Average	Rank
Ravi	90	79	468	78	2

- Name and subject marks → **Data**
- Total, Average, Rank → **Information**

2. Information System

A **system** is a collection of components that work together to complete a task.

An **Information System** is a system that:

1. Receives data (**Input**)
2. Processes data

3. Stores data
4. Provides information (**Output**)

Information System Process

Input → Processing → Storage → Output

Example:

Keyboard → CPU → Memory → Monitor

3. Examples of Information Systems

1. ATM (Automatic Teller Machine)

Input:

- ATM card
- PIN number

Processing:

- Bank computer system processes data

Output:

- Account balance
 - Cash withdrawal details
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2. Fingerprint Attendance System

Input:

- Fingerprint
- Arrival and departure time

Processing:

- Matches fingerprint data

Output:

- Attendance records
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3. QR Code System

- QR codes store information.
 - Scanning a QR code using a smartphone provides additional information from websites.
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4. Business Information System

Data such as:

- Sales
- Purchases
- Income
- Expenses

are processed to obtain:

- Profit
- Loss
- Business reports

Information helps in decision making.

4. Quality Characteristics of Information

Good information should have the following qualities:

Quality	Meaning	Example
Relevancy	Information should be useful for the purpose	Highest educational qualification only
Completeness	All necessary information should be included	Complete survey data
Accuracy	Information should be correct	Correct patient details
Timeliness	Information should be updated	Latest weather report
Cost Effectiveness	Cost of obtaining information should be reasonable	Avoid spending more money than benefit

5. Information and Communication Technology (ICT)

ICT is the technology used to:

- Convert data into information.
- Store information.
- Exchange information between people and systems.

ICT combines:

- Information Technology
 - Communication Technology
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6. Applications of ICT

ICT is used in many fields:

- e-Government
 - Education
 - Health
 - Agriculture
 - Transport
 - Entertainment
 - Business
 - Industries
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7. ICT in e-Government

When the government communicates with citizens and organizations using ICT, it is called **e-Government**.

Examples:

- Government websites
- Online government services
- Digital information portals

Advantages:

- Easy access to government information
 - Faster communication
 - Saves time
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8. ICT in Education

ICT helps teachers and students in learning.

Uses in Classroom

- Presentations
 - Educational videos
 - Creating images and videos
 - Desktop publishing
 - Educational games
 - Internet-based learning
 - CD-ROM learning materials
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Education Anywhere Anytime

Using internet facilities, students can learn from anywhere.

Examples:

- Online learning websites
- Web Based Training (WBT)
- Online courses

Advantages:

- Saves time
 - Reduces travelling costs
 - Flexible learning
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ICT as a Teaching Aid

Teachers can use ICT for:

- Multimedia presentations
 - Animations
 - Audio and videos
 - Lesson preparation
 - Finding educational resources
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9. Learning Management System (LMS)

An **LMS** is a system used to manage teaching and learning activities online.

Services for Students:

- Access learning materials anytime
- Submit assignments
- Participate in forums
- Communicate with teachers
- Join online activities

Services for Schools:

- Manage student information
 - Upload learning materials
 - Monitor student progress
 - Publish results
 - Communicate with parents
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10. Online Distance Learning

Learning through the internet without attending a physical classroom is called **Online Distance Learning**.

Features:

- Flexible learning time
- Digital libraries
- Online assignments and quizzes

- Online teacher discussions
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11. ICT in Health Sector

ICT improves medical services using modern equipment.

Medical Equipment

CAT Scan

- Creates 3D images of body parts.
- Helps diagnose diseases.

MRI Machine

- Uses magnetic fields and radio waves.
- Produces images of internal organs.

ECG Machine

- Records electrical activity of the heart.

EEG Machine

- Records brain activities.

Blood Sugar Testing Machine

- Measures blood glucose level.

Blood Pressure Machine

- Measures blood pressure.
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12. Telemedicine

Using ICT to provide healthcare services to patients who are far away from hospitals is called **Telemedicine**.

Types of Telemedicine:

1. Emergency Telemedicine

- Provides emergency medical advice remotely.

2. Home Health Medicine

- Monitors patients at home.

3. Telemedicine Consultation

- Allows doctors to consult specialists online.

4. Telesurgery

- Doctors perform surgery with remote assistance.

5. Medical Teletraining

- Provides medical training remotely.
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13. ICT in Agriculture

ICT helps farmers improve production.

Applications:

Meteorological Devices

- Provide weather information.

Automated Insect Control

- Detect insect populations.

Field Measuring Devices

- Measure soil conditions.

Drip Irrigation

- Controls water supply.

Automatic Weed Remover

- Removes weeds automatically.

Robotic Seedling Planting

- Plants seedlings using robots.

Robotic Harvesting

- Monitors and harvests crops.

Greenhouse Technology

- Controls temperature, moisture, and light.
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14. ICT in Farm Management

RFID (Radio Frequency Identification)

Used for:

- Identifying animals
- Counting animals
- Tracking locations

Automated Milking Systems

Used for:

- Milking cows
- Checking animal health

CCTV

Used for:

- Farm security
 - Monitoring activities remotely
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15. ICT in Fishing Industry

Sensors placed in the sea provide information about:

- Fish locations
- Fish concentration

Information is sent to fishing boats through the internet.

16. ICT in Manufacturing and Business

ICT helps industries to:

- Produce quality products
- Reduce production costs
- Increase efficiency

Uses of Robots:

- 24-hour operation
- High accuracy

- Less human effort
- Increased efficiency

Examples:

- Car production
 - Packaging
 - Bottle filling
 - Product labeling
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17. ICT in Business

Video Conferencing

Advantages:

- Online meetings
- Saves travelling time and cost

Human Resource Management

Uses:

- Fingerprint attendance
- Employee records
- Salary preparation

e-Banking

Services:

- ATM withdrawals
- Online transactions
- Paying bills
- Checking account balance

Online Shopping

Advantages:

- Available 24 hours
- Buy products from anywhere

- Online payments
 - Home delivery
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18. ICT in Transport

CCTV Systems

Used for:

- Monitoring traffic
- Detecting accidents
- Controlling illegal activities

Traffic Light Control Systems

Used to:

- Control vehicle movement
- Reduce accidents

Parking Identification Systems

Uses sensors to:

- Identify vehicles
 - Control parking access
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19. ICT in Entertainment

ICT is used for:

- Listening to music
 - Watching movies
 - Playing games
 - Social networking
 - Digital photography
 - Reading e-books
 - Watching online videos
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20. Disadvantages of ICT

1. Addiction

- Excessive computer use can affect health and education.

2. Health Problems

- Eye strain
- Back pain
- Headaches
- Obesity

3. Cyber Problems

- Unsafe friendships
- Viruses
- Harmful websites
- Privacy problems

4. Copyright Violations

- Illegal copying of digital content.
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21. Evolution of Computers

Early Computer Development

Abacus (5000 years ago)

- First calculating device.

Pascal's Adding Machine (1642)

- Invented by Blaise Pascal.

Leibniz Machine (1674)

- Improved Pascal's machine.
- Could perform multiplication and division.

Jacquard Loom

- Used punch card technology.

Charles Babbage

- Designed Analytical Engine.
- Known as the **Father of Computing**.

Ada Lovelace

- Considered the **First Programmer**.

Mark I (1944)

- Developed by Howard Aiken.
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22. Generations of Computers

Generation Period		Main Technology
First	1940–1956	Vacuum Tubes
Second	1956–1963	Transistors
Third	1964–1975	Integrated Circuits
Fourth	1975–1989	Microprocessors
Fifth	1989–Present	ULSI, AI, Internet